Spectral Solver Design — Liouville / Sturm–Liouville / Chebyshev Methods for the Stratified Pressure Equation

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0. About this document

This document consolidates four prior technical notes produced during 2026-05-01..03 into a single integrated reference, updated with the post-Phase-0-ext+ conclusions. The four predecessors are:

- `anelastic_SL_spectral_design.md`
- `liouville_SL_spectral_derivation.md`
- `reduced_pressure_liouville.md`
- `liouville_singularity_causality.md`

All four original files are retained in the repository as a record of the reasoning trajectory; each one now carries a prominent header pointing the reader here and to the formal report.

Relationship to the formal report. The formal English report `docs/spectral_stratified_poisson_report_2026-05-` is the citation-ready document (18 pages, authored by Kiriko / Tsinghua University) that presents the Phase 0 ext+ conclusions and quantitative evidence. The present document is the **internal engineering design record**: it retains the full mathematical derivations, the discussion of implementation trade-offs, and the historical context, and is the intended starting point for a developer joining the project.

Three sibling documents:

- `docs/spectral_stratified_poisson_report_2026-05-03.md` — formal English report
- `docs/spectral_experiments.md` — experimental record (Phase 0, reduced-pressure, g-mode, polytropic-index convergence)
- `docs/singular_basis_survey_2026-05-02.md` — GYRE / Dedalus survey

Part I Motivation and Problem Setup

1. Motivation

The existing `pseudo_spectral` solver handles uniform-density, doubly-periodic, two-dimensional incompressible Navier–Stokes ($\rho = \text{const}$, $\nabla \cdot u = 0$) in vorticity–streamfunction form. Stellar convection requires stratified density $\rho_0(y)$, which promotes the pressure equation to a **variable-coefficient elliptic problem**. Fourier modes are no longer eigenfunctions, and the standard $\mathcal{O}(N \log N)$ pseudo-spectral Poisson solve fails.

This document integrates two technical strategies:

1. **Liouville–SL basis (Parts II-III)**. The substitution $\hat{p} = \sqrt{\rho_0}\, q$ reduces the variable-coefficient Poisson operator to a Schrödinger form whose Sturm–Liouville eigenfunctions are common to all k_x. The Poisson solve becomes a pipeline "cuFFT + GEMM + pointwise division + GEMM + cuFFT".
2. **Direct Chebyshev collocation (Part IV)**. The Phase 0 ext+ investigation demonstrates that, for the Eddington standard model ($n = 3$, $\sigma = 3$), raw Chebyshev collocation attains spectral convergence to machine precision without any Liouville-type substitution. **This is the Phase 1 main route**.

Both routes share the same Chebyshev-grid infrastructure; they differ only in how the operator is assembled. The SL route is retained as a Phase 2+ optional backend.

2. Problem formulation

2.1 Variable-density incompressible pressure equation

Dividing the variable-density incompressible momentum equation by ρ_0 and taking the divergence gives

$$\nabla \cdot \left(\frac{1}{\rho_0(y)} \nabla p \right) = f(x) \tag{2.1}$$

where $\rho_0(y)$ is the background stratified density, depending only on y. For the anelastic system $\nabla \cdot (\rho_0 u) = 0$, the same operator appears after the analogous projection; the mathematical structure is identical.

2.2 Fourier reduction in the homogeneous direction

The x direction remains homogeneous and periodic. Expanding $p(x, y) = \sum_{k_x} \hat{p}(k_x, y)\, e^{ik_x x}$ reduces (2.1) for each horizontal mode to a one-dimensional variable-coefficient ODE:

$$\frac{\mathrm{d}}{\mathrm{d}y}\left[\frac{1}{\rho_0(y)} \frac{\mathrm{d}\hat{p}}{\mathrm{d}y} \right] - \frac{k_x^2}{\rho_0(y)}\, \hat{p} = \hat{f}(k_x, y). \tag{2.2}$$

Issue. The $1/\rho_0(y)$ factor in front of the k_x^2 term couples SL modes. A direct eigenfunction expansion of $\frac{\mathrm{d}}{\mathrm{d}y}[(1/\rho_0)\frac{\mathrm{d}}{\mathrm{d}y}]$ alone is not enough: the k_x^2 term introduces dense mode coupling.

The two resolutions of this issue — the Liouville substitution (Parts II-III) and direct Chebyshev collocation (Part IV) — are what this document develops.

Part II Liouville Normal-Form Reduction

3. The Liouville substitution

3.1 Change of dependent variable

Set

$$\boxed{\widehat{p}(y) = \sqrt{\rho_0(y)}\, q(y),} \tag{3.1}$$

and compute the transformed operator.

Proposition 1 (Liouville reduction)**.** *Under the substitution (3.1),*

$$\frac{\mathrm{d}}{\mathrm{d}y}\left[\frac{1}{\rho_0}\frac{\mathrm{d}\widehat{p}}{\mathrm{d}y}\right] = \frac{1}{\sqrt{\rho_0}}\,[q'' + W(y)\,q],$$

*with the **Liouville potential***

$$W(y) \;=\; \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}.$$

Proof. Differentiating $\widehat{p} = \sqrt{\rho_0}\, q$:

$$\widehat{p}' = \frac{\rho_0'}{2\sqrt{\rho_0}}\, q + \sqrt{\rho_0}\, q',$$
$$\frac{1}{\rho_0}\,\widehat{p}' = \frac{\rho_0'}{2\,\rho_0^{3/2}}\, q + \frac{q'}{\sqrt{\rho_0}}.$$

Differentiating the second line:

$$\frac{\mathrm{d}}{\mathrm{d}y}\left[\frac{1}{\rho_0}\,\widehat{p}'\right] = I_1 + I_2,$$
$$I_1 = \left[\frac{\rho_0''}{2\,\rho_0^{3/2}} - \frac{3(\rho_0')^2}{4\,\rho_0^{5/2}}\right] q + \frac{\rho_0'}{2\,\rho_0^{3/2}}\, q',$$
$$I_2 = -\frac{\rho_0'}{2\,\rho_0^{3/2}}\, q' + \frac{q''}{\sqrt{\rho_0}}.$$

The q' contributions cancel, leaving

$$I_1 + I_2 = \frac{1}{\sqrt{\rho_0}}\left[q'' + W(y)\,q\right].$$

□

3.2 The reduced equation

Substituting (3.1) and (3.2) into (2.2) and multiplying by $\sqrt{\rho_0}$:

$$\boxed{q'' + W(y)\,q - k_x^2\, q = g(y), \qquad g \equiv \sqrt{\rho_0}\,\widehat{f}.} \tag{3.4}$$

Defining the **Liouville–Schrödinger operator**

$$\mathcal{T} \equiv \frac{\mathrm{d}^2}{\mathrm{d}y^2} + W(y), \tag{3.5}$$

(3.4) reads

$$\left[\mathcal{T} - k_x^2\right] q = g. \tag{3.6}$$

The key structural property: $\mathcal{T}$ does not depend on k_x. The horizontal wavenumber enters only as an additive shift of eigenvalues. A single precomputed spectral basis therefore serves all k_x simultaneously.

4. Spectral diagonalisation

4.1 The eigenvalue problem

Consider the Sturm–Liouville (equivalently, time-independent Schrödinger) eigenvalue problem

$$\mathcal{T}\,\psi_n \;=\; \psi_n'' + W(y)\,\psi_n \;=\; -\mu_n\,\psi_n, \qquad n = 0, 1, 2, \ldots \tag{4.1}$$

with appropriate boundary conditions (periodic, Dirichlet, or Neumann). Standard SL theory guarantees:

1. Eigenvalues $\{\mu_n\}$ are real and non-decreasing: $\mu_0 \leq \mu_1 \leq \mu_2 \leq \cdots$.
2. Eigenfunctions $\{\psi_n\}$ form a complete orthonormal basis of $L^2([0, L_y])$.
3. When $W \equiv 0$ (uniform density), ψ_n reduces to the standard Fourier modes and $\mu_n = n^2\pi^2/L_y^2$.

Theorem 1 (Universal diagonalisation)**.** *The eigenfunctions $\{\psi_n\}$ simultaneously diagonalise the operator $\mathcal{T} - k_x^2$ for every k_x:*

$$\left[\mathcal{T} - k_x^2\right] \psi_n = -(\mu_n + k_x^2)\,\psi_n.$$

4.2 Solution by eigenfunction expansion

Expanding $q = \sum_n a_n\,\psi_n$ and $g = \sum_n g_n\,\psi_n$ and substituting into (3.6), orthonormality gives

$$\boxed{a_n(k_x) = -\frac{g_n(k_x)}{\mu_n + k_x^2}.} \tag{4.3}$$

No mode coupling. The variable coefficient $1/\rho_0$ is entirely encoded in the eigendata $\{(\mu_n, \psi_n)\}$. Each (k_x, n) solve is a scalar division, structurally identical to the uniform-density Fourier–Poisson solve $\widehat{p}(k) = -\widehat{f}(k)/|k|^2$.

5. Complete SL-GEMM algorithm

5.1 Precomputation (once, or whenever ρ_0 changes)

1. Compute $W(y)$ from $\rho_0(y)$ via (3.3).
2. Solve the 1D Schrödinger eigenvalue problem (4.1) for $\{\mu_n, \psi_n\}$.
3. Form the transform matrix $\Psi_{in} = \psi_n(y_i)$, $\Psi \in \mathbb{R}^{N_y \times N_y}$.
4. Precompute the vector $\sqrt{\rho_0(y_i)}$.

5.2 Per-timestep Poisson solve

Step	Operation	Formula	Cost
1	FFT in x	$f(x,y) \to \hat{f}(k_x,y)$	$\mathcal{O}(N_xN_y \log N_x)$
2	Weight	$g = \sqrt{\rho_0} \cdot \hat{f}$	$\mathcal{O}(N_xN_y)$
3	Forward SL transform	$G = \Psi^\top g$	$\mathcal{O}(N_y^2N_x)$
4	Pointwise divide	$Q_n(k_x) = -G_n/(\mu_n + k_x^2)$	$\mathcal{O}(N_xN_y)$
5	Inverse SL transform	$q = \Psi\, Q$	$\mathcal{O}(N_y^2N_x)$
6	Weight	$\hat{p} = \sqrt{\rho_0} \cdot q$	$\mathcal{O}(N_xN_y)$
7	IFFT in x	$\hat{p}(k_x,y) \to p(x,y)$	$\mathcal{O}(N_xN_y \log N_x)$

Dominant cost: the pair of dense matrix–matrix products in steps 3 and 5, giving overall complexity $\mathcal{O}(N_y^2N_x)$ per Poisson solve.

5.3 Complexity comparison

Method	y direction	Total ($N \times N$)	GPU characteristics
Fourier ($\rho_0 = \text{const}$)	division	$N^2 \log N$	cuFFT (optimal)
Chebyshev + tridiagonal	Thomas alg.	N^2	serial per column
SL-GEMM	**GEMM**	N^3	**cuBLAS batched (tensor cores)**
Iterative (CG / MG)	SpMV $\times k$	N^2k	convergence-dependent

While SL-GEMM has higher asymptotic complexity, dense matrix multiplication achieves near-peak GPU arithmetic throughput (high arithmetic intensity, fully parallel, tensor-core compatible). For $N_y \lesssim 4096$, batched DGEMM wall-clock time is competitive with latency-bound Thomas, and often faster.

Part III Reduced-Pressure Formulation

6. Motivation — a change of dependent variable weakens the singularity

The $1/\rho_0$ coefficient in (2.1) is introduced by the specific algebraic manoeuvre of dividing by ρ_0 before taking the divergence. **Changing the dependent variable changes the singularity**.

Define the **reduced pressure** $\pi \equiv p/\rho_0$ (specific enthalpy perturbation). Substituted into the anelastic momentum equation with the continuity constraint, this yields

$$\nabla \cdot (\rho_0 \nabla \pi) = \tilde{f}, \tag{6.1}$$

where $\tilde{f}$ absorbs the buoyancy, advection, and $\nabla \rho_0$ terms. **The structural change**: the elliptic operator in (6.1) has coefficient ρ_0 (not $1/\rho_0$). The density now appears in the **numerator**.

7. Reduced-pressure Liouville analysis

7.1 Fourier reduction

$$\frac{\mathrm{d}}{\mathrm{d}y}\left[\rho_0 \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}y}\right] - k_x^2 \rho_0 \hat{\pi} = \tilde{f}(k_x, y). \tag{7.1}$$

7.2 Substitution $\hat{\pi} = \rho_0^{-1/2} q$

Direct symbolic computation (verified with SymPy) yields

$$q'' + \widetilde{W}(y)\, q - k_x^2\, q = \tilde{g}, \tag{7.2}$$

with

$$\widetilde{W}(y) = \frac{\rho_0''}{2\rho_0} - \frac{(\rho_0')^2}{4\rho_0^2}. \tag{7.3}$$

Difference from the original potential W: the $(\rho_0')^2/\rho_0^2$ coefficient is $-1/4$ rather than $-3/4$.

7.3 Surface-singularity comparison

For Lane–Emden $n = 3/2$, $\rho_0 \propto (R-y)^{3/2}$:

Formulation	Operator	Surface potential	$\lvert C \rvert$
Original	$\nabla \cdot (\rho_0^{-1} \nabla p)$	$W \approx -21/(16t^2)$	1.3125
Reduced pressure	$\nabla \cdot (\rho_0 \nabla \pi)$	$\widetilde{W} \approx +3/(16t^2)$	0.1875

Singularity strength reduced by a factor of 7, and the sign of C flips from negative (attractive) to positive (repulsive). Frobenius indicial analysis:

Formulation	C	Indicial exponents	Square-integrable?
Original	$-21/16$	$7/4,\ -3/4$	only $\alpha = 7/4$ branch
Reduced pressure	$+3/16$	$3/4,\ 1/4$	**both branches**

This is a qualitative improvement: both linearly independent solutions are square-integrable, and SL theory applies directly.

Scope notice (2026-05-03 update). The factor-of-7 reduction documented above is meaningful only for $\sigma = 3/2$. For the project's principal scenario $\sigma = 3$ (Eddington, where ρ_0 is a polynomial), raw Chebyshev collocation reaches machine precision directly, and this optimisation offers no operational value. See Part IV.

Part IV Direct Chebyshev Collocation (Phase 1 main route)

8. The integer-vs-fractional σ convergence cliff

The central finding of Phase 0 ext+ (2026-05-03): direct Chebyshev collocation of the reduced-pressure operator, with no substitution whatsoever, exhibits a **sharp dichotomy** in convergence behaviour depending on whether the surface exponent σ is integer or fractional.

8.1 Numerical evidence

Solve the manufactured-solution Poisson problem $\rho_0(r) = (1-r)^\sigma$, $\pi_{\text{exact}} = \sin(2\pi r)$, $f = [\rho_0 \pi']' - k_x^2 \rho_0 \pi$ (with f constructed symbolically in SymPy).

N	$\sigma = 3$ error	$\sigma = 3/2$ error (raw)
16	8.5×10^{-8}	2.7×10^{-3}
32	$\sim 10^{-10}$	5.1×10^{-4}
64	6.7×10^{-11}	1.8×10^{-4}
128	$\sim 10^{-9}$ (roundoff)	5.2×10^{-5}
256	3.2×10^{-9}	1.1×10^{-5}

- $\sigma = 3$: exponential convergence to machine precision ($\sim 10^{-10}$ at $N = 64$), followed by a slow roundoff rise — the ideal behaviour of Chebyshev with a polynomial coefficient.
- $\sigma = 3/2$: steady N^{-2} algebraic convergence. No r^α prefactor recovers spectral accuracy.

8.2 Approximation-theoretic explanation (Trefethen Thm 7.2)

The decay of Chebyshev coefficients is governed by the analytic regularity of the function being expanded. For $\rho_0(r) = (1-r)^\sigma$:

- $\sigma \in \mathbb{Z}_{\geq 0}$: ρ_0 is a **polynomial**, its Chebyshev expansion terminates in $\sigma + 1$ terms, and the whole coefficient is captured to machine precision. The spectral convergence of the outer problem is untouched by the surface.
- $\sigma \notin \mathbb{Z}$: $(1-r)^\sigma$ has a **fractional algebraic branch point**. Its Chebyshev coefficients decay only as $N^{-\sigma-1/2}$. L_N approximates L only to this rate, and the rate propagates to the solution — hence N^{-2} for $\sigma = 3/2$.

8.3 Physical significance for stellar structure

The Lane–Emden equation $\theta'' + (2/\xi)\theta' + \theta^n = 0$ with $\theta(\xi_1) = 0$ has surface behaviour $\theta \sim (\xi_1 - \xi)$, so $\rho_0 \propto (\xi_1 - \xi)^n \propto (R - r)^n$. **The polytropic index n is literally the surface exponent σ.**

Index	Physical context	Surface σ	Rate
$n = 1$	outer white-dwarf layer	1	$N^{-3/2}$ algebraic
$n = 3/2$	convective core	$3/2$	N^{-2} algebraic
$n = 2$	main-sequence envelope	2	$N^{-5/2}$ algebraic
$n = 3$	**Eddington radiative model**	**3**	**spectral**
$n = 7/2$	giant hydrogen envelope	$7/2$	N^{-4} algebraic

The Eddington $n = 3$ model is the unique standard physical polytrope in the spectrally-convergent branch. This is a coincidence between "the most-studied historical polytrope" and "the best-behaved Chebyshev regime", and it justifies choosing $n = 3$ as the background for the Phase 1 main route.

9. Chebyshev discretisation

9.1 Chebyshev–Gauss–Lobatto grid

$$\xi_j = \cos\frac{j\pi}{N}, \qquad j = 0, \dots, N, \tag{9.1}$$

mapped affinely to $r \in [a, b]$. Let D denote the Trefethen spectral differentiation matrix of size $(N+1) \times (N+1)$.

9.2 Reduced-pressure operator

With $R_\rho \equiv \text{diag}(\rho_0(r_j))$,

$$L_N = D\, R_\rho\, D - k_x^2\, R_\rho. \tag{9.2}$$

Dirichlet boundary conditions are imposed by strong collocation (unit rows at $j = 0$ and $j = N$, right-hand side set to prescribed values).

9.3 A symmetry caveat

D is **not symmetric** in the standard Euclidean inner product; it is symmetric in the Chebyshev–Gauss–Lobatto quadrature-weighted inner product. Eigenvalue problems must therefore be solved with `numpy.linalg.eig` (LAPACK `geev`), not `eigvalsh`. This was operationally important in Phase 0 ext+ Tests B and C: an initial implementation with `eigvalsh` produced spurious eigenvalues. See `docs/spectral_experiments.md` §19.

Part V Unified Basis Claim — Critical Examination

10. "g-mode as a free by-product" revisited

10.1 The two operators have different singular points

- **Poisson operator**: singular at the **surface** $r = R$ (where $\rho_0 \to 0$). Liouville potential $\widetilde{W} \sim \sigma(\sigma - 2)/[4(R-r)^2]$. The SymPy-derived optimal prefactor is

$$\pi = (R-r)^{\alpha_\star}\, u, \qquad \alpha_\star(\text{Poisson}) = 1 - \sigma/2. \tag{10.1}$$

- **g-mode operator**: singular at the **origin** $r = 0$ (the centrifugal term $\ell(\ell+1)/r^2$). The corresponding prefactor is

$$y_1 \sim r^{\beta_\star}, \qquad \beta_\star(\text{g-mode}) = \ell + 1. \tag{10.2}$$

$\alpha_\star \neq \beta_\star$: **no single set of eigenfunctions simultaneously diagonalises both operators**.

10.2 The correct refined statement

The strong form of the original Liouville programme — "a unified SL basis simultaneously diagonalises the Poisson and g-mode operators" — fails. The weaker, operationally relevant form survives:

- The Chebyshev **mesh** is shared (same CGL nodes).
- The dense-linear-algebra infrastructure (cuBLAS GEMM, VRAM layout) is shared.
- The Poisson solve at each k_x reuses a single LU factorisation (only the scalar shift k_x^2 changes).
- The g-mode calculation is a **separate EVP on the same mesh**, not a free by-product.

The engineering benefit is preserved; the "free mathematical by-product" claim is demoted. This is the central conclusion of Phase 0 ext+ experiments E5 and E6.

Part VI Liouville Singularity — A Physical Interpretation

11. Reversed causality: quantum mechanics vs. astrophysics

The $1/t^2$ behaviour of the Liouville potential as $\rho_0 \to 0$ is mathematically identical to structures in quantum mechanics (the Coulomb potential, the centrifugal barrier). The **causal relationship**, however, is reversed.

11.1 In quantum mechanics: the potential is primary

$$V(r) \text{ (physical input)} \;\longrightarrow\; \psi(r) \to 0 \text{ (mathematical consequence)}$$

A Coulomb potential or centrifugal barrier is the **fundamental physical quantity**. The wavefunction ψ (and hence the probability density $|\psi|^2$) vanishes at certain boundaries *because* the potential forces it to. Techniques such as Coulomb–Sturmian bases ($r^\ell e^{-\alpha r} L_n$), Laguerre DVR, and R-matrix matching are designed around a singularity that **cannot be removed**, because it is a property of the physical system.

11.2 In astrophysics: the density is primary

$$\rho_0(y) \to 0 \text{ (physical input)} \;\longrightarrow\; W(y) \to -\infty \text{ (mathematical artifact)}$$

The density profile $\rho_0(y)$ is determined by hydrostatic equilibrium (Lane–Emden, MESA). The stellar surface is simply where the gas runs out; there is no "infinite potential barrier" in the physics. The original operator $\nabla \cdot (\rho_0^{-1} \nabla p)$ is a **degenerate elliptic** operator at the surface, losing ellipticity smoothly as $\rho_0 \to 0$, without any true singularity. The divergence of W is **manufactured by the $\sqrt{\rho_0}$ substitution** — it is a coordinate singularity, not a physical one.

11.3 Consequences for basis design

1. **Quantum-mechanics methods are applicable but over-engineered.** Coulomb–Sturmian bases were designed for genuine physical singularities; applying them to an artificial coordinate singularity is using a sledgehammer for a thumbtack.
2. **Path A's limitations trace to the same observation.** A prefactor substitution r^α to absorb W's singularity is the astrophysical analogue of the Coulomb–Sturmian $r^\ell e^{-\alpha r}$ factor. There, it is the correct method; here, it attempts to "unwind" a singularity that ought not to exist in the first place.
3. **The natural approach is to return to the original operator.** Reduced pressure (Part III) weakens the singularity from non-integrable attractive to two-branch-integrable repulsive. Raw Chebyshev (Part IV) discretises the coefficient ρ_0 directly, and for polynomial σ encounters no singularity at all.

Summary. Compared with the technical machinery the quantum-mechanics community has developed for genuine Coulomb singularities, the stratified-Poisson problem in astrophysics should be handled with **minimal machinery**. For $\sigma \in \mathbb{Z}$, no special treatment is required; for $\sigma \notin \mathbb{Z}$, a Jacobi-weighted basis (Dedalus) is the only additional machinery needed.

Part VII Phase Roadmap

12. Current status and forward plan

12.1 Completed (Phase 0 ext+, 2026-05-02..03)

- Verified the convergence dichotomy between Lane–Emden $n = 3/2$ and $n = 3$.
- Benchmarked Chebyshev at $N = 48$ (192 DOF) against the GYRE full-gravity 4-variable system: max relative error 1.5×10^{-6}, a $21\times$ reduction in DOF and $350\times$ smaller max error compared with staggered FD at $N_r = 1024$ (4096 DOF).
- Three analytical-ceiling tests (manufactured Poisson, quantum harmonic oscillator, Dirichlet Laplacian) all reach double-precision machine accuracy (10^{-13} to 10^{-15}).
- Verified via barycentric Lagrange that "N coefficients $\neq$ N pixels".
- Chose Path A (raw Chebyshev) as sufficient for $n = 3$.

Full evidence: `docs/spectral_experiments.md` and `docs/spectral_stratified_poisson_report_2026-05-03.m`

12.2 Phase 1: 2D Fourier–Chebyshev Boussinesq

- x **direction**: Fourier (periodic), reusing the `pseudo_spectral` cuFFT infrastructure.
- y **direction**: Chebyshev collocation, GPU-side $D^{(2)}$ by cuBLAS dense GEMM.
- **Physics**: 2D Boussinesq with buoyancy; Poisson via Chebyshev + dense solve.
- **Benchmark**: Rayleigh–Bénard Nu–Ra scaling (Ahlers et al.).
- **Background**: Gaussian transition, moving to Eddington $n = 3$ polytrope.

A design document `docs/phase1_2d_spectral_design.md` will be written when Phase 1 starts in earnest.

12.3 Phase 2: Anelastic upgrade

- Upgrade from Boussinesq to anelastic: $\nabla \cdot (\rho_0 u) = 0$.
- Chebyshev handles the variable-density Poisson directly (raw or reduced-pressure form).
- **SL-GEMM backend as an optional optimisation**: activated only if dense Poisson solves become a GPU-wall-clock bottleneck.

12.4 Phase 3: Live eigenmode projection (differentiator)

- Independent EVP on the same mesh yields g-mode / p-mode eigenpairs.
- Instantaneous flow fields are projected onto the mode basis as a runtime diagnostic.
- **This is the genuine novelty of the project**: convection–pulsation coupling in a 2D nonlinear DNS with live modal projection.

12.5 Phase 4+: Spherical-shell extension

See `docs/sph_spectral_roadmap.md` (long-term).

12.6 Publication strategy

- **Methods paper** (JCP): “Sturm–Liouville spectral methods for stratified astrophysical flows — convergence regimes and GPU implementation”.
- **Applications paper** (A&C / ApJS): “GPU anelastic pseudo-spectral DNS with live eigenmode projection for convection–pulsation coupling diagnostics”.

The applications paper carries the real novelty; the methods paper functions as supporting citation.

Appendix A GPU implementation considerations

A.1 Batched GEMM formulation

Steps 3 and 5 of §5.2 are a single matrix–matrix multiplication:

$$G = \Psi^\top g, \qquad q =,$$

with $g, G, Q, q \in \mathbb{R}^{N_y \times N_x}$. On modern NVIDIA Ampere/Hopper GPUs, FP64 GEMM exceeds 1 TFLOP/s, and the $\mathcal{O}(N)$ arithmetic intensity (flops per byte read) ensures compute-bound execution for $N_y \geq 256$.

A.2 Integration with the time integrator

The SL-GEMM Poisson solver slots into an existing pseudo-spectral time integrator (e.g. IFRK3 + cuFFT) by replacing the spectral-space division $\hat{p} = -\hat{f}/|k|^2$ with the sequence:

cuFFT (R2C in x) $\to$ cuBLAS DGEMM $\to$ pointwise divide $\to$ cuBLAS DGEMM $\to$ cuFFT (C2R in x).

All existing infrastructure — IFRK3 time integration, skew-symmetric convection, 2/3 dealiasing, VRAM frame buffering — is reused without modification.

A.3 Memory footprint

The transform matrix Ψ requires N_y^2 doubles $= 32$ MiB at $N_y = 2048$, which is modest compared with the flow-field arrays ($\sim$ 100 MiB each at 2048^2) and the VRAM frame buffer ($\sim$ 10 GiB).

Appendix B Related methods and literature positioning

B.1 The three-community gap

Community	Representative codes	Radial discretisation	Why SL-GEMM is absent
Stellar pseudo-spectral	ASH, Rayleigh	spherical harmonics (angular) + Chebyshev/FD (radial) + banded	GEMM not competitive on CPU
GPU pseudo-spectral	hit3d, spectralDNS	Fourier in all directions	limited to uniform density
Variable-density GPU CFD	engineering LES codes	finite volume + multigrid	spectral methods rarely used

No existing code combines Liouville reduction, SL spectral basis, GPU GEMM, and anelastic stellar convection.

B.2 SL theory background

- **Liouville normal form**: Sturm 1836, Liouville 1837; see Zettl

(2005) for a modern treatment.

- **SL eigenfunction expansions for elliptic PDEs**: discussed by Boyd

(2001) as theoretically elegant but rarely implemented due to the absence of a fast SL transform.

- **Chebyshev convergence theory**: Trefethen (2013) Ch. 7 (asymptotic coefficient decay for integer vs fractional branch points).
- **Berrut–Trefethen barycentric Lagrange interpolation** (2004): enables a stable, machine-precision evaluation of the spectral representation on any finer grid.

B.3 Relation to GYRE / Dedalus

See `docs/singular_basis_survey_2026-05-02.md` for a detailed survey. The project's novelty positioning, in its final form, is "GPU 2D anelastic DNS with live eigenmode projection" — it does not compete with GYRE (the mature 1D stellar-pulsation code) or Dedalus (the general-purpose spectral-PDE framework) in the 1D pulsation-benchmark arena.

Appendix C Historical record — mapping to the four predecessor documents

Section in this document	Predecessor	Original section
Part I §1-2	anelastic_SL_spectral_design.md	§1-2
Part II §3-4	liouville_SL_spectral_derivation.md	§3-4
Part II §5	liouville_SL_spectral_derivation.md §5 + anelastic_SL_spectral_design.md §5	
Part III §6-7	reduced_pressure_liouville.md	entire
Part IV §8-9	polytropic_index_spectral_convergence_2026-05-03.md + formal report §3	
Part V §10	new (Phase 0 ext+ E5/E6 conclusions)	
Part VI §11	liouville_singularity_causality.md	entire
Part VII §12	new (post Phase 0 ext+ roadmap)	
Appendix A	anelastic_SL_spectral_design.md §8 + liouville_SL_spectral_derivation.md §8	
Appendix B	anelastic_SL_spectral_design.md §7 + liouville_SL_spectral_derivation.md §9	

All four predecessor files are retained in the repository as a record of the reasoning trajectory. Each carries a header update block pointing the reader to this document and to the formal report; new work should proceed from this document and the formal report only.

0. About this document

This document consolidates four experimental notes from 2026-05-02..03 into a single integrated record, reorganised around **setup / result / interpretation** for each experiment. The original four notes remain in the repository as an archival reasoning trajectory:

- `docs/anelastic_sl_phase0_2026-05-02.md` (§1-2 here: SL Poisson feasibility)
- `docs/reduced_pressure_experiments_2026-05-02.md` (§3-5 here: reduced-pressure validation)
- `docs/gmode_experiments_2026-05-02.md` (§6-15 here: the full g-mode experiment chain, Exp A through Exp K)
- `docs/polytropic_index_spectral_convergence_2026-05-03.md` (§16-19 here: the σ-dichotomy study)

Reproducibility protocol. Every experiment freezes its expected numerical output as an `EXPECTED` constant in the corresponding script. A `python scripts/<name>.py --verify` invocation must exit zero to confirm the numbers in the tables below. Any intentional change to these numbers must be landed in the same commit that updates both the `EXPECTED` and the corresponding table.

Sibling documents:

- `docs/spectral_solver_design.md` — design, derivations, roadmap.
- `docs/spectral_stratified_poisson_report_2026-05-03.md` — formal English report.
- `docs/singular_basis_survey_2026-05-02.md` — GYRE / Dedalus survey.

Part I SL Poisson feasibility (Phase 0, 2026-05-02)

1. Lane-Emden background and the Liouville potential W

1.1 Setup

The Emden equation $\theta'' + (2/\xi)\theta' + \theta^n = 0$ is integrated to the first zero ξ_1.

1.2 Numerical results for Lane-Emden $n = 3/2$

Quantity	Value
ξ_1 (polytrope radius)	3.653754
ρ/ρ_c range	$[1.0,\ 2\times 10^{-5}]$
$W(y)$ full range	$[-1.35\times 10^{6},\ -3.3]$
Surface region $\rho < 0.01$	$\lvert W\rvert_{\max} = 1.35\times 10^{6}$ (divergent)
Truncated domain $r/R_\star \in [0, 0.94]$ ($\rho > 0.01$)	$W \in [-398, -3.3]$, bounded

Script: `scripts/anelastic_sl_phase0.py`.

1.3 Conclusion

The surface singularity exists as anticipated at the design stage. For $n = 3/2$ the safe truncation is $r/R_\star \leq 0.94$ (the convective interior, excluding the thin surface atmosphere). This is the standard ASH / Rayleigh prescription.

2026-05-03 update: the truncation is required only for $n = 3/2$. For $n = 3$ (Eddington), $\rho_0 \propto (R - r)^3$ is a polynomial and Chebyshev handles it to machine precision without truncation (see Part IV).

2. SL eigenproblem + Fourier-limit degeneracy test

2.1 Discretisation

Interior finite differences on nodes $1, \ldots, N - 2$ with implicit Dirichlet endpoints, $N = 512$. `scipy.sparse.linalg.eigsh` extracts the lowest 256 eigenvalues.

2.2 Fourier limit verification ($W = 0$)

With $W = 0$ the SL eigenvalues should reduce to $(n\pi/L)^2$.

n	μ_{SL}	$(n\pi/L)^2$	rel. err.
1	11.146	11.146	3.1×10^{-6}
20	4452.8	4458.4	1.26×10^{-3}

The $n = 20$ relative error matches the FD stencil theoretical limit $n^4\pi^4\Delta y^2/(12L^4) \approx 1.26 \times 10^{-3}$ exactly. Measured $=$ theory $\Rightarrow$ discretisation verified.

2.3 End-to-end SL-Poisson manufactured solution

N_{modes}	err_{L^2}	improvement
5	3.00×10^{-2}	–
10	7.34×10^{-3}	4.1×
20	1.37×10^{-3}	5.4×
40	2.18×10^{-4}	6.3×
80	3.27×10^{-5}	6.7×
160	5.99×10^{-6}	5.5×
256	3.74×10^{-6}	converged to FD floor

The convergence slope $\log(\mathrm{err})$ vs. $\log(N)$ is ≈ -3.5, i.e. **algebraic**, not exponential. This is consistent with the theoretical expectation that the truncated surface $\rho \to 0$ should degrade the SL expansion from exponential to algebraic convergence.

2.4 Sturm oscillation + Tassoul asymptote (Phase 0 extensions E1, E2)

- **E1 (Sturm).** Each ψ_n has exactly n internal zeros (21 of 21 tested). This verifies bit-level correctness of the eigenvectors and preservation of topology.
- **E2 (Tassoul).** ΔP_n asymptotes to a constant with std/mean $= 8 \times 10^{-4}$. The shape matches Tassoul (1980) qualitatively, but a factor of 2.9 in magnitude remains — traceable to the Cowling slab approximation (which drops the $\ell(\ell+1)/r^2$ centrifugal term).

2.5 Convergence order vs. cutoff (E3) — the smooth-vs-singular contrast

Scanning the Lane-Emden density threshold $\rho_{\text{threshold}}$:

cutoff	r_{hi}	err(256)	slope
0.1	0.77	4.3×10^{-7}	-2.42
0.01	0.94	3.7×10^{-6}	-2.39
0.001	0.99	2.7×10^{-5}	-2.26
0.0001	0.997	1.5×10^{-4}	-1.78

The closer to $\rho = 0$, the slower the convergence — the singularity's influence is quantitatively visible.

Gaussian-capped smooth $\rho(y) = \exp(-2y^2) + 0.05$ **(no singularity)**: $\text{err}(256) = 8.3 \times 10^{-7}$, semilog slope -0.049, giving err $\sim \exp(-0.05\,N)$ — **exponential convergence**.

Publication-level summary:

> "The SL method is exponentially convergent for smooth stratification. Algebraic convergence observed on Lane-Emden polytropes is entirely attributable to the surface singularity $\rho(R_\star) = 0$, quantifiable via cutoff scaling analysis."

2.6 Brunt–Väisälä $N^2(r)$ vs. Liouville $W(r)$ — physical division of labour (E4)

Quantity	Encoding	Purpose
$W(r)$	pure density stratification (ρ'')	SL Poisson Liouville potential
$N^2(r)$	density + temperature (Schwarzschild)	g-mode frequencies / convective stability

Finding. Lane-Emden with γ-adiabatic law is Schwarzschild-neutral; $N^2 \equiv 0$ identically. A non-adiabatic perturbation ($\delta = 0.1 \sin(2\pi r)$ on top of T) gives $|N^2| \sim 1$, comparable in scale to $|W| \sim$ 10-400.

2.7 Phase 0 cumulative verification table

Check	Status	Strength
Lane-Emden $W(y)$ singularity localisation	PASS	quantified via cutoff
SL discretisation Fourier limit	PASS	matches FD theoretical limit
Stable solution of first 256 eigenpairs	PASS	scipy eigsh, < 1 s
SL-Poisson $\mathrm{err}_{L^2} = 3.7 \times 10^{-6}$	PASS	engineering-ready
Sturm oscillation (E1)	PASS	21/21 bit-correct
Tassoul asymptotic ΔP (E2)	PASS	asymptote attained
Exponential vs. algebraic (E3)	PASS	smooth $\Rightarrow$ exponential confirmed
$N^2 \leftrightarrow W$ division of labour (E4)	PASS	Phase 2/3 scope clarified

Phase 0 gate was originally PASS. Phase 0 ext+ subsequently demoted the "SL as optimal basis" angle to "same-mesh independent EVP" (see `docs/spectral_solver_design.md` Part V).

Part II Reduced-pressure follow-up (2026-05-02)

The experiments in this part numerically validate the theoretical prediction of `docs/reduced_pressure_liouville.md` — namely, that the reduced-pressure formulation weakens the surface singularity by a factor of 7. **Scope**: meaningful only for Lane-Emden $n = 3/2$ (fractional σ); not applicable to the project's main $n = 3$ scenario (see Part IV).

3. Experiment A — breaking the Chebyshev floor

Script: `scripts/reduced_pressure_chebyshev.py`.

3.1 Motivation

The parent report's §9 discretised the SL eigenvalue problem with finite differences, giving a floor $\sim 10^{-7}$ set by FD accuracy. A Chebyshev collocation eigensolver for (μ_n, ψ_n) breaks this floor and makes the two formulations quantitatively distinguishable.

3.2 Result (Lane-Emden $n = 3/2$, cutoff 0.01, N-mode scan)

N	err (original)	err (reduced-p)	ratio
5	6.0×10^{-3}	3.7×10^{-4}	$16\times$
10	8.4×10^{-4}	6.9×10^{-5}	$12\times$
20	7.8×10^{-5}	7.6×10^{-6}	$10\times$
40	5.1×10^{-6}	5.7×10^{-7}	$9\times$
80	3.1×10^{-7}	1.5×10^{-7}	$2\times$
256	1.4×10^{-7}	1.5×10^{-7}	$1\times$ (FD floor)

Reduced-pressure is $10\times$ more accurate in the spectral range $N \leq 40$; at high N both hit the FD floor.

3.3 Interpretation

- **Consistent $10\times$ advantage at low mode count.** GPU GEMM cost scales as N_y^2; using 20-40 SL modes is the natural engineering target, and this is exactly the range where reduced pressure excels.
- **Advantage vanishes for a smooth ρ.** With a Gaussian density profile (no singularity), the two formulations give identical convergence curves — confirming that the improvement stems from the weaker singularity, not from any generic property of the reduced-pressure variable.
- **The repulsive potential is better conditioned.** The original strongly attractive potential ($C = -21/16$) distorts low-order ψ_n toward the singular boundary; compensating requires many high-order modes. The weakly repulsive reduced-pressure potential ($C = +3/16$) leaves low-order ψ_n closer to Fourier modes, enabling rapid convergence at small mode count.

4. Experiment B — end-to-end Poisson convergence (on π, not q)

Script: `scripts/reduced_pressure_poisson_end2end.py`.

4.1 Setup

Manufactured $\pi_{\text{exact}}(x, y) = \sin(2\pi k_x x) \sin(\pi(y - y_{\text{lo}})/L)$ with $k_x = 2$. Forward and backward transport verified.

4.2 Principal result ($\rho_{\text{cut}} = 0.01$)

$\text{err}_{L^2}(\pi) = 1.8 \times 10^{-6}$ at $N = 256$ modes.

4.3 k_x-independence verification (Experiment C)

A single precomputed (μ_n, ψ_n) set attains comparable accuracy for every $k_x \in \{1, 2, 4, 8, 16\}$, confirming the theoretical prediction of §5.

5. Symbolic correction to the $\widetilde{W}$ coefficient

The parent report's eq. (12) had a sign error (a missing factor of 3). The SymPy script `scripts/reduced_pressure_liouvi` computes, for $n = 3/2$,

$$\frac{1}{\sqrt{\rho_0}} \frac{\mathrm{d}}{\mathrm{d}y} \left[\rho_0 \frac{\mathrm{d}}{\mathrm{d}y} \left(\frac{q}{\sqrt{\rho_0}} \right) \right] = q'' + \frac{3}{16\, t^2} q,$$

with $t = R - y$. This confirms $C = +3/16$, in agreement with §3. The parent report has been corrected.

Part III g-mode infrastructure and validation (Exps A-K)

The following is a condensed walk-through of the full g-mode experiment chain, from the initial incompressible-buoyancy simplification to the GYRE-compatible 4-variable adiabatic operator. Each entry is organised as **setup / result / conclusion**. Scripts are located in `scripts/gmode_exp_*.py`, with shared infrastructure in `scripts/gmode_infra.py`.

6. Exp A — Lane-Emden $\widetilde{W}$-proxy heuristic

Script: `gmode_exp_a_lane_emden.py`, commit `8aa3476`.

Setup. Lane-Emden $n = 3/2$, $\rho_{\text{cut}} = 0.05$, cavity $r \in [0.15, 0.844]$. Use $N^2_{\text{proxy}} \equiv -\widetilde{W}(r)$ as input to the Cowling solver (30 radial orders, last-5 tail average).

Result. $\Delta P_{\text{tail}}/\Delta P_{\text{Tassoul}} = 0.852$ (within the acceptance window 0.80-1.20, PASS).

Conclusion. A pure pipeline smoke test. Lane-Emden itself is isentropic with $N^2 = 0$ and supports no true g-modes, but `solve_gmode_cowling` should reproduce the Tassoul asymptote whenever any positive N^2 profile is fed in.

7. Exp B — artificial Gaussian-bump N^2 convergence

Script: `gmode_exp_b_stratified.py`, commit `8aa3476`.

Setup. Artificial Gaussian $N^2(r)$ on $[0.2, 1.0]$, $r_c = 0.6$, $\sigma = 0.2$, with a $\sin^2$ taper. Resolution scan $N_r \in \{256, 512, 1024, 2048\}$.

Result ($N_r = 2048$). $\Delta P_{\text{tail}}/\Delta P_{\text{Tassoul}} = 0.9993$; $|\text{ratio} - 1| = 7.5 \times 10^{-4}$. Convergence rate $\mathcal{O}(N_r^{-2})$, matching the second-order FD expectation.

Conclusion. The **first genuine g-mode calculation in the repository**. The infrastructure is ready for MESA profile ingestion, multi-cavity / radiative-convective boundary extensions, etc.

8. Exp C — Chebyshev collocation g-mode solver

Script: `gmode_exp_c_chebyshev.py`, commit `e703991`.

Setup. Same Gaussian bump as Exp B, now discretised with Chebyshev collocation (`solve_gmode_cowling_cheb`).

Result ($N_{\textbf{Cheb}} = 512$). $|\text{ratio} - 1| = 6.85 \times 10^{-5}$ — $4\times$ smaller DOF and $10\times$ smaller error than Exp B at $N_r = 2048$.

Spurious-mode guard. `n_modes = max(10, N_\text{Cheb}//5)` is enforced to avoid spectral-tail pollution. The Chebyshev D^2 is non-symmetric in the standard inner product, so one must use `numpy.linalg.eig` rather than `eigvalsh`. This lesson reappears in Phase 0 ext+ Tests B and C.

9. Exp D — polytropic profile via a MESA-style parser

Script: `gmode_exp_d_polytrope_profile.py`, commit `c8b655c`.

Setup. A MESA-style column-table reader ingests a synthetic $n = 3$ polytrope with a Gaussian N^2 fixture (600 rows), then hands it to the Cowling solver.

Result. Chebyshev $N = 512$ gives $|\text{ratio} - 1| = 8.1 \times 10^{-5}$. Drift relative to the in-memory Exp B/C numbers is 18%, dominated by the 600-row fixture interpolation error.

Conclusion. The parser pipeline is clean and can be swapped for a real `profile*.data`.

10. Exp E-G — 2-variable anelastic operator (transition phase)

Scripts: `gmode_exp_e_anelastic_linop.py`, `gmode_exp_f_variable_rho.py`, `gmode_exp_g_spherical_scalar.p`

Setup. A 2-variable (y_1, p') anelastic operator; eliminating p' yields a scalar reduction. E/F/G cross-validate three algebraic forms.

Result. Exp E PASS on the Gaussian-bump cavity with the wide 0.85-1.20 acceptance window. Exp F PASS with variable density. Exp G spherical scalar vs. 2-variable PASS with $\mathcal{O}(N_r^{-2})$ convergence.

2026-05-02 corrections log. §7 originally described the scalar solver as the "Boussinesq limit of the 2-variable operator" — this was wrong. The scalar solver is the **slab / local-Cartesian** approximation (drops $\ell(\ell + 1)/r^2$), not a thermodynamic truncation (Boussinesq vs. anelastic). Consequently, the B-G PASS verdicts demonstrate **internal consistency**, not **external correctness**. The first external benchmark, Exp H, exposes this.

11. Exp H — first GYRE benchmark (exposing the problem)

Script: `gmode_exp_h_gyre_benchmark.py` plus `gmode_exp_h_run_gyre.sh`.

Setup. Build GYRE in the MESA SDK environment, run it on the bundled Lane-Emden $n = 3$ `poly3` case, and compare against the Python 2-variable anelastic solver under full gravity.

Result. $n_g = 1$ ratio 2.2× (120% disagreement); $n_g = 5$ ratio 1.3; $n_g = 10$ ratio 1.1; high-n_g ratio tends to 1 (the Boussinesq limit).

Diagnosis. Not a bug — a physics mismatch. `solve_anelastic_2var` uses only ρ_0, N^2 and **silently drops the V, U, Γ_1 coupling**. GYRE's $\alpha_{\text{grv}} = 0$ pure Cowling 2-variable system requires five structure coefficients $V, U, A^\star, c_1, \Gamma_1$. Our solver is a Boussinesq-like simplification, not "anelastic in the stellar-oscillation sense".

Conclusion. Re-implement the operator to match the GYRE equations (Exp I / Exp J).

12. Exp I — 2-variable Cowling GYRE-compatible (first external benchmark)

Script: `gmode_exp_i_gyre_compat.py`, commit `953d49f`.

Setup. Implement GYRE's $\alpha_{\text{grv}} = 0$ 2-variable Cowling equations, with $(y_1, y_2) = (x^{2-\ell}\xi_r/r, x^{2-\ell}P'/(\rho g r))$ and the five structure coefficients $V, U, A^\star, c_1, \Gamma_1$ read from GYRE's `poly3.txt`. Staggered FD at $N_r = 1024$, cutoffs inner_cut $= 0.01$, outer_cut $= 0.999$.

Result (vs. GYRE Cowling):

n_g	$\omega^2_{\text{GYRE(Cow)}}$	ω^2_{ours}	rel. diff.
1	2.85195	2.85195	2.3×10^{-6}
2	1.36145	1.36146	3.1×10^{-6}
5	0.37473	0.37472	2.1×10^{-5}
10	0.11850	0.11843	5.6×10^{-4}

Max relative difference 5.6×10^{-4}, far below the 1% target. $n_g = 1$ **agrees to 5-6 significant figures.** The first apples-to-apples external benchmark PASS.

Cowling-limit quantified. GYRE Cowling vs. GYRE full differ by $\sim 13\%$ at $n_g = 1$ — the known Cowling approximation error (Unno et al. 1989).

13. Exp J — 4-variable full-gravity GYRE-compatible (FD production reference)

Script: `gmode_exp_j_full_gyre_compat.py`, commit `be94af9`.

Setup. Lift Cowling to $\alpha_{\text{grv}} = 1$ full 4-variable system (see `docs/spectral_solver_design.md` §4.2 and the formal report §4.2), adding $y_3 = \Phi'/(gr)$ and $y_4 = (\mathrm{d}\Phi'/\mathrm{d}r)/g$. Same FD grid.

Two bookkeeping bugs caught during implementation. An initial version gave 2.5% rel. diff. at $n_g = 1$ — physically untenable. Re-reading `A_t.inc` (GYRE stores the Jacobian transpose, so `A_t(i, j) = A(j, i)`) identified two errors:

1. Eq. 1 was missing the $\lambda/(c_1\omega^2)\, y_3$ term (present only when $\alpha_{\text{grv}} = 1$). This inverse-ω^2 term is what couples Φ' back into the displacement equation in the full-gravity case.
2. Eq. 2 had the y_3 coefficient as $-A^\star$ instead of 0, and was missing the $-y_4$ term.

After correction the residual drops by four orders of magnitude:

n_g	$\omega^2_{\text{GYRE(full)}}$	$\omega^2_{\text{ours(full)}}$	rel. diff.
1	2.51593	2.51593	5.9×10^{-7}
2	1.28571	1.28571	2.7×10^{-5}
5	0.36993	0.36992	2.0×10^{-4}
10	0.11807	0.11801	5.3×10^{-4}

$n_g = 1$ **agrees to 6 significant figures.** This is the frozen FD production reference; its `--verify` regression is the oracle for any future CUDA port.

14. Exp K — Chebyshev 4-variable (spectral production)

Script: `gmode_exp_k_chebyshev_full.py`, commit `0da140f`.

Setup. Same equations and boundary conditions as Exp J, now discretised by Chebyshev collocation on $x \in [0.01, 0.999]$. `load_gyre_structure_interp_cheb` uses `scipy.interpolate.CubicSpline` (an initial implementation using `numpy.interp` hit a 3×10^{-5} floor; swapping to a cubic spline lowered it by four orders of magnitude to 8.7×10^{-9}).

Result ($N = 48$, 192 DOF):

n_g	ω^2_{GYRE}	$\omega^2_{\text{Chebyshev}}$	rel. diff.
1	2.51593	2.51593	5.9×10^{-7}
2	1.28571	1.28571	2.7×10^{-5}
5	0.36993	0.36992	2.0×10^{-4}
10	0.11807	0.11801	5.3×10^{-4}

Same accuracy with $21\times$ **fewer DOF and** $350\times$ **smaller maximum error** compared with Exp J at $N_r = 1024$.

15. Production-readiness summary

Operator	Method	$n_g = 1$ err vs. GYRE full
‘solve$_{g}mode_{c}owling$\`$(slab)$	FD, Boussinesq-like	$\sim 220\%$ (educational only)
‘solve$_{g}mode_{c}owling_{s}pherical$\`	FD, scalar reduction	120%
‘solve$_{a}nelastic_{2}var$\`	FD, no $V/U/\Gamma_1$	120%
‘solve$_{g}mode_{c}owling_{g}yre_{c}ompat$\`$(2-var)$	FD, Cowling	13.4% (Cowling limit)
‘solve_gmode_full_gyre_compat‘ (4-var)	FD, full gravity	5.9×10^{-7} (FD production ref)
‘solve_gmode_full_chebyshev‘ (Exp K)	Chebyshev, full	5.9×10^{-7} (spectral production, $21\times$ less DOF)

Part IV Polytropic-index convergence dichotomy

Background. Phase 0 (Part I) observed $N^{-2.4}$ algebraic convergence on Lane-Emden. A systematic Phase 0 ext+ scan reveals that the convergence order depends sharply on whether the surface exponent σ is integer or fractional, with clear consequences for the project's choice of background polytrope.

16. E6 v2 convergence scan (SymPy-forced manufactured solution)

Script: `scripts/spectral_liouville_convergence_v2.py`.

16.1 Setup

$\rho(r) = (1-r)^\sigma$ on $[0,1]$; $\pi_{\text{exact}} = \sin(2\pi r)$ (Dirichlet-compatible); the forcing $f = [\rho\pi']' - k_x^2\rho\pi$ is constructed symbolically in SymPy and evaluated on the CGL grid. Dirichlet BCs $\pi(0) = \pi(R) = 0$.

16.2 $\sigma = 3$ (Lane-Emden $n = 3$)

N	err (raw)	err ($\alpha = 1 - \sigma/2 = -1/2$)
16	8.5×10^{-8}	2.4×10^{-3}
32	$\sim 10^{-10}$	$\sim 10^{-4}$
64	6.7×10^{-11}	1.5×10^{-4}
128	$\sim 10^{-9}$ (roundoff)	$\sim 10^{-5}$
256	3.2×10^{-9}	9.1×10^{-6}

Raw Chebyshev reaches machine precision. Adding a prefactor makes the error **worse** (fractional α introduces algebraic irregularity at the endpoint).

16.3 $\sigma = 3/2$ (Lane-Emden $n = 3/2$)

N	err (raw)	$\alpha = 1/4$	$\alpha = -1/2$	$\alpha = -3/4$
16	2.7×10^{-3}	1.1×10^{-2}	1.3×10^{-2}	7.8×10^{-2}
64	1.8×10^{-4}	7.5×10^{-4}	1.1×10^{-3}	2.1×10^{-2}
256	1.1×10^{-5}	4.8×10^{-5}	8.3×10^{-5}	5.4×10^{-3}

$N^{-2.0}$ **algebraic convergence for all** α**.** No simple power-law prefactor restores spectral accuracy.

17. Why integer vs. fractional σ matters

17.1 Approximation-theoretic view

The decay of Chebyshev coefficients is governed by analytic regularity (Trefethen 2013, Thm. 7.2):

- $\rho(r) = (R-r)^3 = R^3 - 3R^2r + 3Rr^2 - r^3$ is a **polynomial**, with a finite Chebyshev expansion (4 terms). The smoothness of $\rho(r)\pi(r)$ is inherited entirely from $\pi(r)$.
- $\rho(r) = (R-r)^{3/2}$ is analytic on $[0, R)$ but has a branch point at R. Its Chebyshev coefficients decay only as $N^{-\sigma-1/2} \sim N^{-2}$.

The N^{-2} convergence is a direct consequence of Chebyshev's inability to resolve a fractional-power branch point with exponential accuracy.

17.2 Physical significance (Lane-Emden surface structure)

The Lane-Emden equation $\theta'' + (2/\xi)\theta' + \theta^n = 0$ with $\theta(\xi_1) = 0$ has surface behaviour $\theta(\xi) \sim (\xi_1 - \xi)$, so $\rho \propto (\xi_1 - \xi)^n \propto (R-r)^n$.

The polytropic index n is literally the surface exponent σ. See the table in `docs/spectral_solver_design.md` Part IV §8.3.

18. What actually works for fractional σ

1. **Jacobi-weighted basis** (Dedalus). Expand π in $\{(1-r)^\sigma J_n^{(\sigma,0)}(r)\}$ — the basis carries the singular behaviour, and the coefficient expansion is over polynomials. **Gives spectral convergence for any $\sigma > -1$.**
2. **Coordinate stretching** (Kosloff–Tal-Ezer). Introduce a change of variable $r = r(s)$ that concentrates integration near the surface. Used neither by GYRE nor by Dedalus.

Both options lie **outside the Liouville framework**.

19. Analytical ceiling tests (E7b)

Script: `scripts/spectral_analytical_ceiling.py`.

Motivation. Does the 8.7×10^{-9} floor of Exp K represent the Chebyshev ceiling itself, or is it a property of the GYRE input data? Analytical-solution tests discriminate.

19.1 Test A — manufactured Poisson (reused from §16)

$\sigma = 3$ with SymPy-exact forcing. Reaches 5.5×10^{-13} at $N = 24$ — essentially double-precision rounding.

19.2 Test B — quantum harmonic oscillator on $[-10, 10]$

$-\psi'' + x^2\psi = \lambda\psi$, exact eigenvalues $\lambda_n = 2n + 1$. The first five eigenvalues reach 3×10^{-13} relative error at $N = 64$.

19.3 Test C — Dirichlet Laplacian on $[0, 1]$

$-u'' = \lambda u$, exact eigenvalues $\lambda_n = n^2\pi^2$. $N = 16$ already reaches 10^{-14}.

19.4 A key lesson — non-symmetry of D^2

An initial implementation of Tests B and C with `np.linalg.eigvalsh` gave divergent / spurious eigenvalues. The Chebyshev D^2 is not symmetric in the Euclidean inner product; one must use `np.linalg.eig` and filter for finite, positive, real-part eigenvalues. The lesson is in the literature but easy to forget.

19.5 Conclusion

Exp K's 8.7×10^{-9} floor is not a spectral limit. It is the precision ceiling of GYRE's 999-point `poly3.txt` input. Recovering more precision would require rebuilding the polytrope with a GL6 integrator at $\sim 10^4$ nodes and relative tolerance 10^{-14}.

Part V Barycentric interpolation — resolution vs. representation

20. The spectral representation is a continuous function

Script: `scripts/spectral_resolution_demo.py`.

20.1 Concept

The $N + 1$ Chebyshev coefficients define a **continuous function** evaluable at any point by barycentric Lagrange interpolation (Berrut & Trefethen 2004, SIAM Rev 46, 501-517):

$$u(r^\star) = \frac{\sum_j w_j u_j / (r^\star - r_j)}{\sum_j w_j / (r^\star - r_j)}, \quad w_j = (-1)^j c_j,$$

with $c_0 = c_N = 1/2$, $c_j = 1$ otherwise. $\mathcal{O}(N)$ per evaluation point, stable against catastrophic cancellation, error $\leq 10^{-12}$.

20.2 Numerical verification

The Exp K $N = 48$ (49 CGL nodes) representation, barycentric-evaluated at 4096 points, vs. the Exp J FD $N_r = 1024$ solution cubic-spline interpolated at the same 4096 points: max diff 3.4×10^{-3}, **exactly the Exp K discretisation error**; the barycentric contribution is $< 10^{-12}$.

20.3 Implication for 2D simulations

A pseudo-spectral 2048^2 field can be **rendered at 4K / 8K resolution without loss**. The real resolution ceiling is the 2/3 dealiasing cutoff ($2N/3 \approx 1365^2$), not the grid itself.

Part VI Frozen regression assets

21. Frozen 1D g-mode operators

Operator	Method	Accuracy vs. GYRE	Role
`solve_gmode_full_gyre_compat` (`gmode_infra.py`)	Staggered FD $N_r = 1024$	$n_g = 1$ 5.9×10^{-7}, max 5.3×10^{-4}	FD regression oracle
`solve_gmode_cowling_gyre_compat` (`gmode_infra.py`)	Staggered FD $N_r = 1024$	vs. GYRE Cowling 5.6×10^{-4}	FD Cowling baseline
`solve_gmode_full_chebyshev` (`gmode_exp_k_chebyshev_full.py`)	Chebyshev $N = 48$	max 1.5×10^{-6}	**Chebyshev production ref.**

22. Analytical-verification scripts (frozen `EXPECTED`)

- `gmode_exp_i_gyre_compat.py --verify` — 2-var Cowling, max 5.6×10^{-4}.
- `gmode_exp_j_full_gyre_compat.py --verify` — 4-var full, max 5.3×10^{-4}.
- `gmode_exp_k_chebyshev_full.py --verify` — Chebyshev 4-var, max 1.5×10^{-6}.
- `spectral_analytical_ceiling.py` — three ceiling tests, all 10^{-13} to 10^{-15}.

23. Exploratory scripts (retained, not to be modified)

- `gmode_exp_a..g_*.py` — educational baselines (Boussinesq-like simplifications).
- `gmode_exp_h_*.py` — the first GYRE benchmark, diagnostic record.
- `reduced_pressure_*.py` — Liouville-form validation suite.
- `anelastic_sl_phase0*.py` — Lane-Emden $n = 3/2$ Phase 0.
- `spectral_liouville_beta_derivation.py` — SymPy $\alpha_\star$.
- `spectral_liouville_convergence_v2.py` — σ-dichotomy data.
- `spectral_liouville_prefactor.py` — Path A α sweep.
- `spectral_resolution_demo.py` — barycentric demo.

Appendix A Five central findings of Phase 0 ext+ (F1-F5)

The following five points, surfaced during the Phase 0 ext+ conversation, extend beyond the tabulated experimental data:

- **F1.** Exp K's 8.7×10^{-9} floor is not a spectral limit but the GYRE 999-point input precision ceiling. Verified by the analytical-ceiling tests.
- **F2.** "N coefficients $\neq N$ pixels." The $N+1$ Chebyshev coefficients define a continuous function evaluable at arbitrary resolution via barycentric Lagrange interpolation.
- **F3.** Gibbs vs. stellar pulsation. Low-frequency linear problems reach machine precision with spectral methods; strong-shock / supersonic flows do not (switch to `cart_ale2`). For the stellar2d use case (pulsation + weak-to-moderate convection + smooth polytropic background), the practical spectral ceiling is far above 2D FD.
- **F4.** Revised project positioning. Novelty is not in 1D stellar pulsation (GYRE / Reese–Lignières / Dedalus occupy that territory) but in **2D GPU DNS with live eigenmode projection**.
- **F5.** The strong form of "unified basis simultaneously diagonalises Poisson and g-mode" fails ($\alpha_\star \neq \beta_\star$); what survives is "same-mesh independent EVP".

Appendix B Mapping to the four predecessor documents

Section in this document	Predecessor	Original section
Part I §1-2	anelastic_sl_phase0_2026-05-02.md	entire
Part II §3-5	reduced_pressure_experiments_2026-05-02.md	entire
Part III §6-14	gmode_experiments_2026-05-02.md	§2-15
Part III §15	gmode_experiments_2026-05-02.md	§15 production readiness
Part IV §16-19	polytropic_index_spectral_convergence_2026-05-03.md + phase0_ext_plus_summary_2026-05-03.md	F1, §16
Part V §20	phase0_ext_plus_summary_2026-05-03.md	F2 (resolution)
Part VI §21-23	phase0_ext_plus_summary_2026-05-03.md	frozen assets
Appendix A	phase0_ext_plus_summary_2026-05-03.md	F1-F5

All four predecessor files remain in the repository as a reasoning trajectory. Each carries an update block pointing here and to the formal report; new work should be framed against this document and the formal report.